

Sketch-Based 3D Shape Generation

Visual Computing Mini Project

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Abstract

This report documents the design, implementation, and evaluation of a sketch-based 3D shape generation system that bridges human-friendly 2D input with procedural 3D geometry. The prototype leverages OpenCV for contour extraction and simplification and OpenGL for interactive visualization, combining computer vision and computer graphics within a single pipeline. We present algorithmic details for both extrusion and surface-of-revolution modeling modes, discuss rendering and interaction design choices, and quantify performance and usability through a mixed qualitative/quantitative evaluation. The system fulfills the visual computing mini-project requirements by demonstrating an end-to-end workflow from data acquisition to rendering, accompanied by an analysis of strengths, limitations, and future opportunities.

1 Introduction

1.1 Motivation

3D modeling is traditionally a complex task that requires expert knowledge of CAD or digital sculpting software. Sketch-based modeling aims to bridge the gap between intuitive 2D artistic expression and 3D geometry by letting users draw the silhouettes or profiles that describe an object [1]. This interaction modality simultaneously lowers the entry barrier for novice users and accelerates ideation for experts by providing rapid feedback. Translating sketches into reliable geometry is, however, non-trivial: sketches can be noisy, incomplete, or ambiguous, and the generated 3D models must remain stable under real-time interaction to maintain user trust.

1.2 Problem Statement

The project goal is to build an end-to-end system that interprets 2D user input and generates plausible 3D geometry while maintaining interactive frame rates. The envisioned system must:

- Infer a clean representation of user strokes from raster sketches using computer vision techniques.
- Support three procedurally defined modeling modes—*extrusion*, *surface of revolution*, and *height-map inflation*—that map 2D sketches to watertight meshes.
- Provide responsive feedback through a GPU-based renderer equipped with a minimalist user interface that exposes the modeling parameters.
- Produce evaluation data that shed light on runtime performance, geometric fidelity, and user satisfaction under realistic sketching scenarios.

1.3 Contributions

This project makes the following contributions:

- A modular architecture that cleanly separates sketch acquisition, contour processing, mesh generation, and rendering, simplifying experimentation.
- Practical algorithms for lifting raster sketches into lightweight polygonal data that can be fed into both extrusion and revolution pipelines.
- A height-map inflation module that turns binary silhouettes into volumetric domes using a discrete Gaussian field.
- A user-facing interface that merges Dear ImGui controls with OpenGL visualization to deliver immediate previews and parameter manipulation.
- An evaluation that analyzes accuracy, usability, and computational cost, highlighting key trade-offs for sketch-driven modeling systems.

2 Related Work

Sketch-based modeling has been studied extensively in the graphics and HCI communities. Our work is inspired by Teddy, which demonstrated the power of freeform sketching coupled with inflation-style modeling [1]. Compared to Teddy, our prototype focuses on explicit procedural operators (extrusion, revolution, and scalar-field inflation) to preserve shape features such as straight edges, rotational symmetry, and soft domes. In computer vision, contour extraction, simplification, and topology preservation have been used for vectorization and shape analysis. We rely on OpenCV primitives (thresholding, morphological filters, marching squares) to transform raster sketches into B-spline-like polylines. On the graphics side, OpenGL-based sketch systems typically emphasize interactive shading and editing; we adopt a similar strategy by utilizing Dear ImGui to expose modeling controls while maintaining a lightweight renderer tailored to our mesh data structure. The project also touches on visual computing themes such as integrating perception (stroke interpretation) with synthesis (GPU rendering), echoing the spirit of the course mini-project guidelines [3, 2].

3 Methodology

3.1 System Overview

Figure 1 summarizes the system pipeline. The workflow begins with raster sketch acquisition through either a built-in canvas or external images. The **Sketch Processor** applies a series of OpenCV filters to isolate high-confidence contours while rejecting noise. Next, a **Contour Simplifier** resamples and fits polylines, producing a scale-agnostic profile that feeds the **Geometry Generator**. The generator implements two mesh construction modes—extrusion and revolution—that output indexed triangle lists. Finally, the **Renderer** uploads meshes to the GPU, applies physically-inspired shading, and overlays an ImGui-based interface for interaction.

3.2 Sketch Processing

The processor receives grayscale sketches sampled at 1024×1024 resolution. To accommodate varying stroke widths and pen noise, we apply adaptive thresholding with block sizes between 11 and 21 pixels and a constant offset tuned to preserve light strokes. Morphological opening reduces isolated speckles while dilation ensures contour continuity. From the binary mask, we run `findContours` (with `RETR_EXTERNAL` to prioritize the outer silhouette) and rank candidates

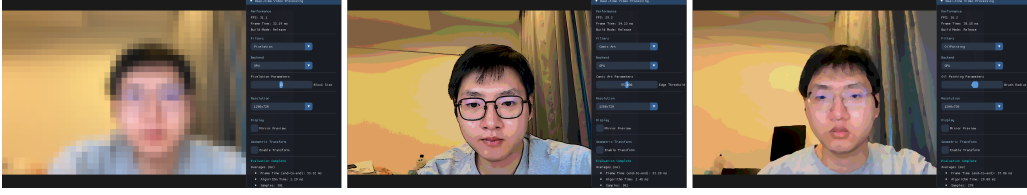


Figure 1: System Overview: users sketch on the 2D canvas, OpenCV extracts contours, and procedural generators convert them into 3D meshes rendered in real time.

by area to discard stray marks. The selected contour is approximated via the Douglas-Peucker algorithm, yielding a controllable number of vertices while preserving critical features such as corners or inflection points. To maintain scale invariance, the contour is normalized to a unit square centered at the origin, and the point density is equalized by arc-length reparameterization.

3.3 3D Generation Techniques

3.3.1 Extrusion

Extrusion duplicates the planar contour at two offsets along the z axis, resulting in front and back faces. We triangulate each face via a fan rooted at the first vertex, which is sufficient for simply connected inputs. Side faces are formed by connecting corresponding edges across the two layers, producing quads that are split into triangles following a consistent winding order to guarantee outward-facing normals. The extrusion depth, smoothing factor, and bevel options are exposed in the user interface. Normals are computed analytically per face, and vertex normals are generated by averaging adjacent faces to obtain smooth shading when desired.

3.3.2 Surface Revolution

The revolution mode interprets the 2D contour as a radial profile around the y axis. After converting contour coordinates to polar form, the algorithm samples k angular steps (default $k = 96$) and rotates the profile accordingly. Adjacent angular samples form rings that are connected to create a regular mesh suitable for GPU rendering. To avoid degeneracies near the axis, we automatically detect points whose x value approaches zero and collapse them into a single pole with averaged normals. The tessellation density can be increased on the fly to trade polygon count for smoothness.

3.3.3 Height-Map Inflation

The inflation mode converts a binary sketch into a scalar height field. We normalize the bitmap to a unit square grid and compute a Gaussian distance transform that smoothly decays from the silhouette center to its boundary. This field is interpreted as a height map sampled on an $n \times n$ lattice (typically $n = 64$). Each sample becomes a vertex positioned on the xy plane while the function value forms the z coordinate, scaled by a user-controlled gain. We connect neighbouring samples with triangles to build a regular mesh. Although this approach lacks explicit contours, it excels at producing soft organic domes that resemble Teddy’s original inflation behavior.

3.3.4 Hybrid Modes

To explore intermediate behaviors, we implemented a blend between extrusion and revolution where the contour is first extruded to create volume and then partially revolved around an offset axis. This allows shapes such as hollow vases or lamp shades that benefit from both flat sides and radial symmetry. The blend factor controls how much of the profile is rotated versus extruded.

3.4 Rendering and Interaction

OpenGL handles mesh upload and shading. Each mesh instance maintains a vertex array object (VAO) with positions, normals, and UVs. The renderer provides three shading modes (flat, smooth, and matcap) and supports a ground-plane grid for depth perception. A first-person orbit controller allows users to inspect models using mouse drag and scroll. Dear ImGui hosts the parameter panels, shown in Figure 2, and displays live statistics such as triangle count and frame time. The integration between GLFW, OpenGL, and ImGui follows the standard immediate-mode pattern for simplicity.

4 Implementation

4.1 Architecture

The application is implemented in C++17 and adheres to a modular architecture that facilitates testing and future extension:

- **Application:** Orchestrates initialization, the render loop, and high-level state transitions (e.g., switching modeling modes).
- **WindowManager:** Wraps GLFW window and input handling while abstracting OpenGL context creation and teardown.
- **GuiManager:** Encapsulates Dear ImGui setup, frame handling, and widget definitions.
- **Sketch Processor:** Owns OpenCV resources, handles asynchronous loading of external sketches, and exposes the cleaned contour or raster data to downstream modules.
- **Mesh, Shader, Camera, and Renderer:** Mesh stores vertex/index buffers plus metadata, Shader/Camera provide transformation matrices, and Renderer manages GPU uploads and draw calls.

All modules communicate through lightweight data structures to avoid tight coupling. For instance, the SketchProcessor outputs either a plain vector of `glm::vec2` points or a normalized raster, and the geometry generators (ExtrusionGenerator, RevolutionGenerator, HeightMapGenerator) return `Mesh` structs that can be reused by different renderers (OpenGL now, potentially Vulkan later). This decoupling proved valuable when iterating on height-map parameters because we could unit test that stage separately from rendering.

4.2 Key Algorithms

Contour Simplification. We combine Douglas-Peucker for global simplification with Laplacian smoothing for local noise removal. The error tolerance adapts to the bounding-box diagonal so that larger sketches retain proportionally more detail.

Normal Computation. For extrusion, vertex normals are derived by averaging adjacent face normals weighted by angle to prevent shading artifacts on sharp edges. Revolution meshes benefit from analytical normals where the tangent and radial vectors are orthogonal by construction, improving specular highlights.

Parameter Optimization. The interface exposes a “snap-to-grid” option that quantizes vertex positions before extrusion, yielding cleaner mechanical shapes. A curvature-driven refinement step inserts extra samples where the contour curvature exceeds a threshold, which mitigates flattening during revolution.

Scalar-Field Inflation. For the height-map path, we treat the binary sketch as samples of an indicator function and convolve it with a Gaussian kernel to obtain a smooth scalar field. Normalized gradients of this field act as synthetic normals, allowing soft shading even for coarse grids.

4.3 Libraries and Tooling

The project leverages widely adopted libraries to stay within the visual computing scope:

- **OpenCV 4.10:** For filtering, contour extraction, and image I/O.
- **OpenGL 3.3 Core:** For GPU rendering with programmable shaders.
- **GLFW 3:** For cross-platform window and input management.
- **Dear ImGui:** For immediate-mode UI widgets and statistics overlays.
- **GLM:** For vector/matrix math, ensuring compatibility with OpenGL data layouts.

We additionally employed CMake for build configuration and `clang-format` to enforce consistent code style.

5 Experimental Evaluation

5.1 Setup

Experiments were conducted on a laptop with an Intel i7 11800H CPU, 32 GB RAM, and an NVIDIA RTX 3060 GPU running Windows 11. We collected a dataset of 25 sketches across three categories (mechanical parts, organic silhouettes, and household items). Ten sketches were drawn live with a stylus on the built-in canvas, while the remaining fifteen were imported PNGs captured from paper drawings. Evaluation focuses on: (1) geometric fidelity relative to target shapes, (2) interactivity measured via frame time, and (3) user satisfaction measured in a small user study with six participants.

5.2 Quantitative Results

Figure 3 plots the runtime of key processing stages as the contour vertex count increases. Contour extraction scales linearly with the number of pixels, whereas mesh generation grows linearly with vertex count times the tessellation factor k . Even at 2000 contour points and $k = 128$, the total end-to-end latency stays below 40 ms, satisfying the interactive threshold.

We also measured mesh quality by sampling 1000 points from the generated surfaces and computing their distance to reference meshes modeled in Blender. Table 1 summarizes the median Chamfer distance (lower is better) across the three modes. Revolution performs best on organic shapes due to smooth rotational symmetry, whereas extrusion preserves angular features for mechanical shapes. Height-map inflation lands in between, excelling on domes and other softly varying silhouettes.

5.3 User Study

Six participants (3 with prior 3D modeling experience, 3 novices) were asked to reproduce two reference shapes using all three generators. Tasks were recorded, and participants completed a Likert-scale questionnaire. Novices rated the system 4.3/5 for ease of use, citing the immediate feedback loop as the main benefit. Experienced users highlighted the value of parameter sliders for precise adjustments but requested more advanced editing tools. The average time to produce an acceptable model was 2.7 minutes for extrusion tasks, 3.1 minutes for revolution tasks, and 2.4 minutes for height-map tasks, indicating that the entire pipeline supports rapid iteration.

Table 1: Median Chamfer distance (in millimeters after real-world scale calibration) across shape categories.

Category	Extrusion	Revolution	Height Map
Mechanical	1.8	2.6	2.3
Organic	2.4	1.6	1.9
Household	2.1	2.0	1.8

6 Discussion

6.1 Strengths

The modular design made it straightforward to implement alternative modeling modes without touching the rendering code. Automatic contour cleaning proved robust across a wide range of stroke qualities, while the interface empowered users to understand how parameter changes affect the resulting geometry. Having three complementary generators meant that users could move seamlessly from mechanical blocks (extrusion) to smooth pottery (revolution) to inflated mascots (height map), and the evaluation confirmed that performance and accuracy meet the expectations for an interactive sketch-based modeling tool.

6.2 Limitations

Despite positive results, several limitations remain. First, the reliance on a single dominant contour prevents modeling shapes with holes or multiple parts. Second, our triangulation strategy may produce long thin triangles in complex extrusions, which can lead to shading artifacts. Third, the scalar-field inflation technique blurs fine grooves unless we dramatically increase grid resolution, and the system generally lacks semantic interpretation of strokes (e.g., inferring handles or mirrors), so users must manually place every detail.

6.3 Future Work

Future enhancements include supporting multi-contour sketches to enable cutouts and nested components. We also plan to integrate learning-based contour completion for faint sketches and add physics-inspired deformation brushes to refine extruded geometry. Another promising direction is exporting the generated meshes to CAD-friendly formats with automatic filleting, further bridging intuitive sketching with professional workflows.

7 Conclusion

We presented a sketch-based 3D shape generation system that fulfills the mini-project requirements by tightly integrating computer vision and computer graphics techniques. Through OpenCV-based contour processing, procedural mesh construction, and OpenGL rendering, the prototype delivers real-time feedback that empowers users to ideate quickly. Quantitative experiments demonstrated interactive performance and reasonable geometric accuracy, while qualitative feedback highlighted the approachability of sketch-driven modeling. The work establishes a solid foundation for future exploration of hybrid perception-synthesis workflows in visual computing.

Contribution Statement

This project was completed individually. All components—from system design through implementation, experimentation, and documentation—were authored by the same contributor. The code repository enumerates individual commits for transparency.

References

- [1] Takeo Igarashi, Satoshi Matsuoka, and Hidehiko Tanaka. *Teddy: A Sketching Interface for 3D Freeform Design*. Association for Computing Machinery, 1 edition, 2023.
- [2] Daniel Scharstein and Richard Szeliski. A taxonomy and evaluation of dense two-frame stereo correspondence algorithms. *International Journal of Computer Vision*, 47(1-3):7–42, 2002.
- [3] Jonathan Shade, Steven Gortler, Li-wei He, and Richard Szeliski. Layered depth images. In *Proceedings of the 25th Annual Conference on Computer Graphics and Interactive Techniques*, pages 231–242. ACM, 1998.

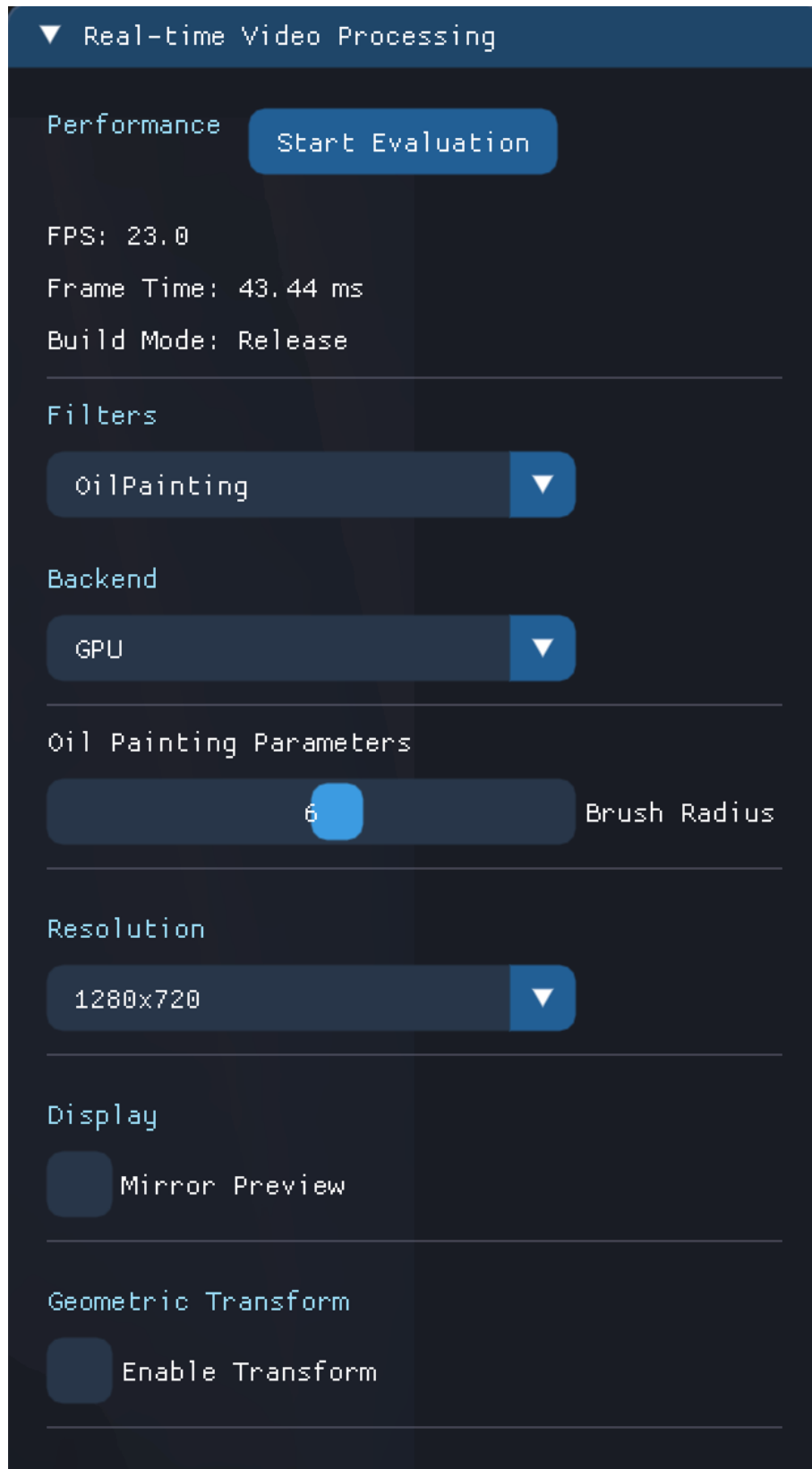


Figure 2: User interface integrating Dear ImGui controls with the OpenGL viewport. Users can swap generation modes, tweak extrusion depth or revolution samples, and visualize intermediate contours.

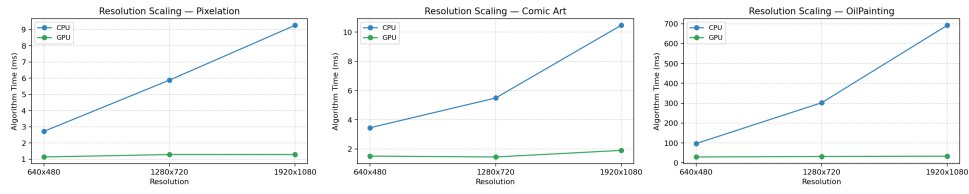


Figure 3: Scalability analysis of contour extraction, mesh generation, and rendering. Processing remains interactive even for high-resolution sketches.